Mineralium Deposita (2000) 35: 672–682 © Springer-Verlag 2000

ARTICLE

D. T. A. Symons · T. B. Symons · D. F. Sangster

Paleomagnetism of the Society Cliffs dolostone and the age of the Nanisivik zinc deposits, Baffin Island, Canada

Received: 9 January 1999 / Accepted: 3 March 2000

Abstract Dolostones of the ~1200 Ma Society Cliffs Formation within the hydrothermal zone surrounding the Nanisivik zinc deposits retain a stable characteristic remanent magnetization (ChRM) on alternating field and thermal step demagnetization. Based on the thermal data and saturation isothermal remanence analyses, the ChRM resides in pseudosingle domain magnetite and hematite. A paleomagnetic fold test favours a post-folding ChRM, and a paleomagnetic contact test, using a Franklin gabbro dike, indicates that the ChRM predates ~720 Ma. The pole position calculated from the ChRM direction is at 168.2°E, 42.8°N ($\delta_p = 4.9°$, $\delta_m = 6.8°$), giving an age of 1095 ± 10 Ma on the well-defined "Logan Loop" portion of the North American apparent polar wander path. This age is considered to date recrystallization of the dolostone host rocks in the halo around the hydrothermal sulfide deposits. No evidence is found for a postulated Cretaceous remagnetization event in the region.

Introduction

The original aim of this study, initiated in 1989, was to date the Nanisivik Mississippi Valley-type (MVT) zinc deposits (Fig. 1). The ore, however, was found to contain too much pyrite to be successfully demagnetized. Sulfides oxidized to magnetite at ~400 °C on thermal step demagnetization before the remanence of the primary magnetite could be isolated in the 450 to 585 °C range, and the skin effect prevented current penetration to permit isolation of the primary remanence by alternating field (AF) demagnetization. However, the remanence of the host Society Cliffs dolostones was defined and recent evidence suggests that all of the sampled dolostone sites were well within the zone of hydrothermal activity that accompanied ore deposition.

D. T. A. Symons (✉) · T. B. Symons
Earth Sciences, School of Physical Sciences,
University of Windsor, Windsor, Ontario,
N9B 3P4, Canada
e-mail: dsymons@uwindsor.ca
Tel.: +1-519-2533000, ext. 2493; Fax: +1-519-9737081

D. F. Sangster
2335 Russvern Dr., North Gower,
Ontario, K0A 2T0, Canada

Geology

Overview

The geology of the Nanisivik sulfide deposits on northern Baffin Island, Canada, has been described in several publications and recently summarized by Sutherland and Dumka (1995; Fig. 2). This brief description is mainly to provide evidence that limits the age of the deposits.

The deposits occur in laminated to massive light grey dolostones of the Society Cliffs Formation (Table 1). It is the lower formation of the Uluksan Group, a carbonate platform sequence in the middle of the >600-m-thick Mesoproterozoic Bylot Supergroup. The supergroup rests unconformably on an Archean to Paleoproterozoic basement complex of granites, migmatites and gneisses of the amphibolite to granulite metamorphic facies that give radiometric ages of >1700 Ma (Knight and Jackson 1994). The Bylot Supergroup was deposited in the Borden rift basin (Fig. 1), a graben that formed within the pre-existing Canadian Shield.

At the Nanisivik mine (Figs. 2 and 3), the Bylot Supergroup sediments are tilted at 10 to 20° northerly in horst and graben fault blocks that step down northwards towards the Milne Inlet trough on the north side of Strathcona Sound on a series of normal faults. Several features indicate periodic movement on these northwest-trending faults from sometime before Bylot Supergroup deposition to the present (Jackson and Morgan 1978; Jackson and Iannelli 1981). These include: (1) coincidence of the faults with recrystallized

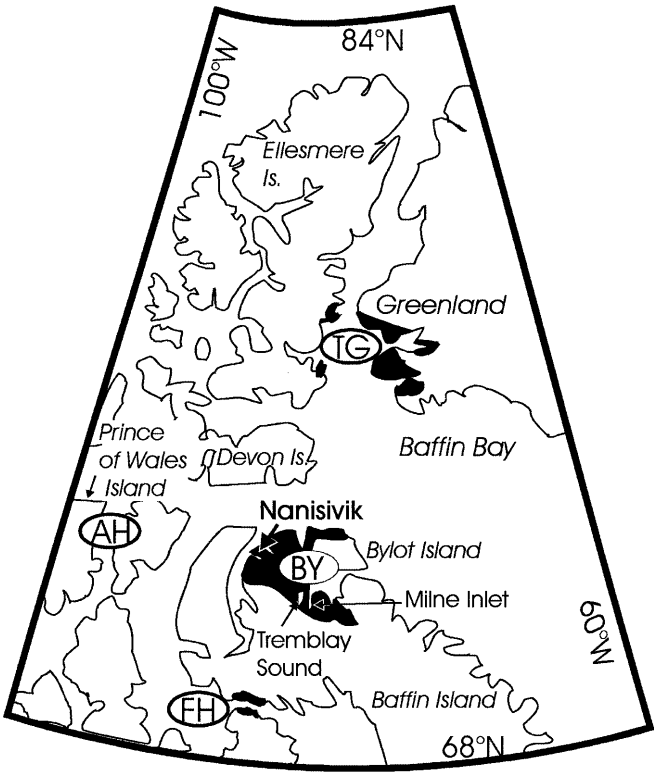


Fig. 1 Location of Nanisivik in the Canadian Arctic Archipelago. *Black fill* indicates the outcrop areas for the Bylot Supergroup (*BY*) and coeval Aston and Hunting formations (*AH*), Fury and Hecla Group (*FH*) and Thule Group (*TG*)

mylonite zones in the Paleoproterozoic basement; (2) differential offsets in the Bylot Supergroup showing syndepositional displacements; (3) intrusion of a dike that belongs to the ~720 Ma Franklin diabase dike swarm along a fault; (4) displacements of early Paleozoic and Cretaceous–Eocene strata by more than 1100 m on these faults; and (5) minor displacements of Recent glacial moraines and beaches.

Existing absolute dates

Several ages have been published for the Bylot Supergroup. The Nauyat basalts, at its base in the Eqalulik Group (Table 1), have yielded 17 K–Ar whole rock ages that are as old as 1221 Ma, with a mean of 946 Ma, and a Rb–Sr whole rock errorchron of 1129 Ma based on data from 6 of 14 analytical samples (Jackson and Iannelli 1981; Jackson et al. 1990; Knight and Jackson 1994). These basalts are deemed to be correlative with the Savage Point sill in the Aston Formation on Prince of Wales Island (Knight and Jackson 1994) and with the Goding Bay sill in the Thule Supergroup on southeastern Ellesmere Island (Fig. 1) that have both given U–Pb zircon/baddeleyite ages of 1268 Ma (LeCheminant and Heaman 1991). These sills also give paleomagnetic poles that agree with those of the Mackenzie dike swarm, which is well dated by U–Pb zircon/baddeleyite at 1267 ± 2 to 1272 ± 3 Ma (Fahrig et al. 1981; LeCheminant and Heaman 1991). Thus, the onset of Bylot Supergroup deposition occurred at ~1270 ± 3 Ma. Further, L.C. Kah and F. Marcantonio (personal communication, 1999) have determined a consistent Pb–Pb age of 1198 ± 24 Ma (2σ) for Society Cliffs dolostones on Bylot Island, and an average age of 1204 ± 22 Ma for the deposition of the Society Cliffs dolostones and the Victor Bay and Athole Point limestones at Tremblay Sound. These dates set ~1200 ± 20 Ma as the maximum age for the sulfide deposition at Nanisivik.

Some paleomagnetic data are available for the Bylot Supergroup on Baffin Island. Fahrig et al. (1981) have reported characteristic remanent magnetic (ChRM) directions for five basalt flows in the basal Nauyat Formation and for seven sites in purple sandstones of the overlying Adams Sound Formation, which together give a pole at 167°W, 1°N ($\delta_p = 3°$, $\delta_m = 6°$) (Fig. 4). This pole coincides with a cluster of poles from the ~1270 Ma Mackenzie igneous event, and it confirms that deposition of the supergroup began at this time. In addition, they reported data from five sites in maroon sandstones from the basal Aston Formation on Prince of Wales Island that also fall in the cluster.

Fahrig et al. (1981) also measured samples from massive red sandstone units in a section of grey–green silty sandstones of the Strathcona Sound Formation about 30 km north of Nanisivik. They obtained 17 normally and 5 reversely polarized sites that defined a pole at 156°W, 8°N ($\delta_p = 2°$, $\delta_m = 4°$). Using the then accepted date of ~1220 Ma for the Mackenzie igneous event and of ~1100 Ma for the Logan dikes to the south in the Canadian Shield around Lake Superior, Fahrig et al. (1981) estimated that the Strathcona Sound sandstones were deposited ~16.5 m.y. after ~1220 Ma and that the entire Bylot Supergroup sedimentary rock sequence was deposited in ~18 m.y., assuming constant rates of polar wander. Updating the argument, and scaling crudely between the well-defined ~1270 Ma cluster of Mackenzie igneous event poles and ~1105 to ~1109 Ma "Logan Loop" cluster of poles (Symons et al. 1994), it is suggested that: (1) Bylot Supergroup deposition began at ~1270 Ma, (2) the Strathcona Sound Formation was deposited at ~1230 Ma, (3) significantly for this study, the Society Cliffs carbonates that host the Nanisivik ore were deposited at ~ 1250 Ma, and (4) Bylot Supergroup deposition ended at ~1225 Ma.

The minimum age for the mineralization is defined geologically by a diabase dike, the Mine Dike, that cuts and alters the ore in the Main Zone of the Nanisivik deposit (Figs. 2 and 3). The dike is one of an extensive swarm of undeformed and unmetamorphosed subvertical dikes that strike northwest to north-northwest throughout Baffin Island and the adjacent mainland (Fahrig and West 1986). Based on paleomagnetic studies (Fahrig et al. 1971; Fahrig and Schwarz 1973), the dikes were assigned to the Franklin igneous event, now well dated elsewhere by U–Pb methods at 723 + 4/−2 to 718 ± 2 Ma (Heaman et al. 1992). Later, scattered

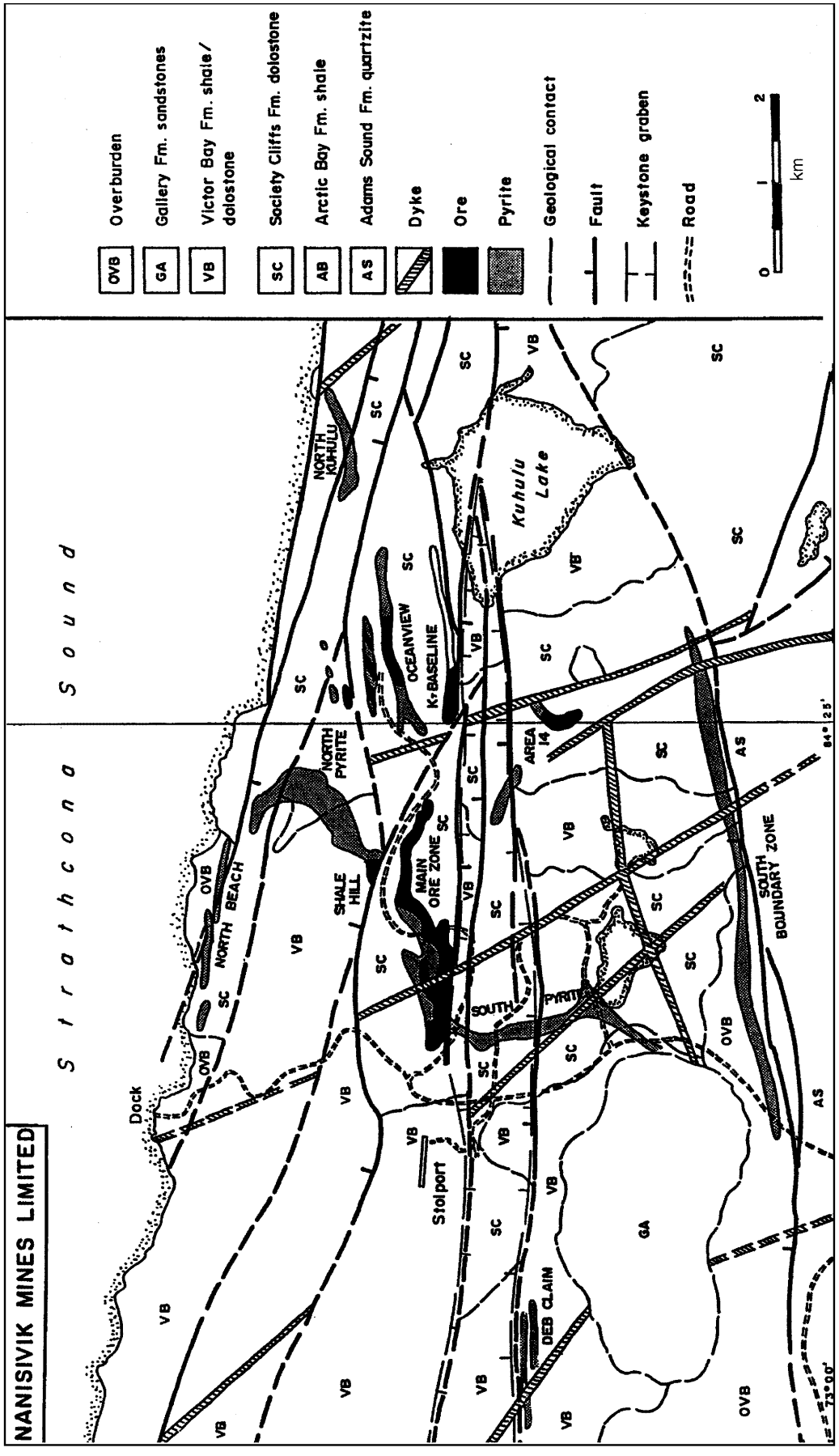


Fig. 2 Geology of the Nanisivik mine area. (From Sutherland and Dumka 1995)

Table 1 Stratigraphic section for the Nanisivik area

Lower Ordovician/Middle–Upper Cambrian, ~500+/−20 Ma	
Admiralty Group	
Turner Cliffs Fm.	Dolostone, sandstone
Gallery Fm.	Sandstone
Unconformity	
Neoproterozoic	
Franklin diabase dikes, ~720 Ma	
Intrusive contact	
Mesoproterozoic ~1120 (?) Ma–1270 Ma	
Bylot Supergroup	
Nunatsiaq Gp. (upper clastic sequence)	
Elwin Fm.	Sandstone
Strathcona Sound Fm.	Sandstone, shale, siltstone
Uluksan Gp. (carbonate platform sequence)	
Victor Bay Fm.	Shale, dolostone
Society Cliffs Fm.	Dolostone, 1200 ± 20 Ma
Equalulik Gp. (lower clastic sequence)	
Arctic Bay Fm.	Shale, siltstone
Adams Sound Fm.	Quartz sandstone
Nauyat Fm.	Basalt, sandstone, 1270 ± 3 Ma
Nonconformity	
Archean–Paleoproterozoic basement	Metamorphosed at ~2500 Ma
	Gneiss, granite and migmatite of amphibolite to granulite facies

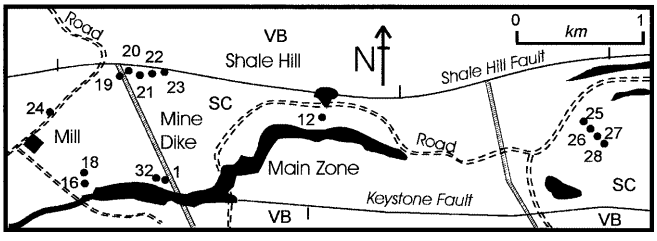


Fig. 3 Location of the sampling sites close to the Main Zone and deposit at Nanisivik. Legend as in Fig. 2

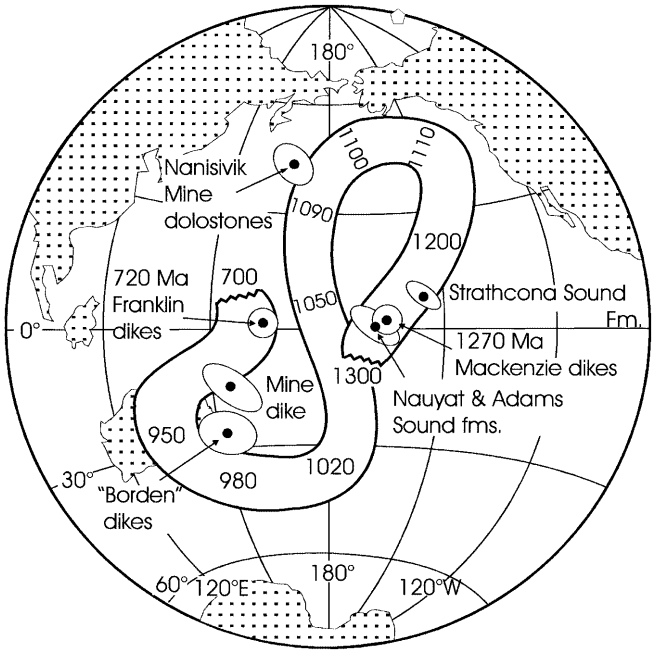


Fig. 4 North American apparent polar wander path for 1300 to 700 Ma. The path is taken from summaries published by: Irving et al (1972) and LeCheminant and Heaman (1989) for the 1270 ± 3 Ma Mackenzie igneous event; Fahrig et al (1981) for 1270 to 1200 Ma; Ernst and Buchan (1993) for 1200 to 1110 Ma; Symons et al (1994) for 1110 to 1050 Ma; Dunlop et al (1985) for 1050 to 720 Ma; and Palmer and Hayatsu (1975) and Heaman et al (1992) for the 720 ± 3 Ma Franklin igneous event. Poles from the Baffin Island area are shown with their cone or oval of 95% confidence

K–Ar radiometric data and additional paleomagnetic work resulted in the northwest-trending dikes being assigned to the ~720 Ma Franklin event and the north-northwest-trending dikes to the ~950 Ma Borden event (Christie and Fahrig 1983; Fahrig and West 1986). Recently, however, two "typical" Borden dikes, ~150 km southeast of Nanisivik, have yielded U–Pb baddeleyite ages of 716+4/−5 Ma and 720 ± 8 Ma, confirming that all of the dikes belong to the Franklin event (Pehrsson and Buchan 1999), and setting the minimum mineralization age at ~720 Ma.

Mineralization

Massive, fissure-fill and disseminated sulfides exist within dolostones of the Society Cliffs Formation at several localities on northern Baffin Island (Olson 1984) as well as in the overlying Victor Bay and underlying Arctic Bay formations. At Nanisivik about 100 Mt of sulfide mineralization have been found within an area of 12 by 8 km (Fig. 2), of which about 85 Mt is barren massive pyrite (Sutherland and Dumka 1995; R. Sutherland, personal communication, 1998). Approximately 15 Mt of sphalerite and galena ore, grading ~10% Zn and ~1% Pb, occurs in the central part of the mineralized area, north of the Keystone fault (Figs. 2 and 3). The largest of the six known zinc-bearing zones at Nanisivik is the Main Zone or Main Lens (Figs. 2 and 3). All of the ore is hosted by the dolostones of the Society Cliffs Formation as irregular, commonly sinuous, tabular sheets and passages with occasional highly discordant vertical zones, referred to as "keels", that extend as much as ~70 m below the sheets. The Main Zone is a sinuous lenticular body, cutting across strata dipping 15° to the north and west, with an horizontal upper contact that varies little in elevation along its 3000 m length. The ore is coarse-grained and comprises pyrite (after marcasite), sphalerite, dolomite and galena, possibly with trace amounts of wurtzite. Paragenetic studies indicate that these ore stage minerals were precipitated concurrently (Olson 1984). Although silver grades of as much as 250 g/t have been reported, no silver-bearing mineral has been identified (Ford 1986; Sutherland and Dumka 1995).

In the sampling programme of 1989, the authors did not appreciate the size of the mineralized area and the sampled dolostone sites (Fig. 3) were thought to be

sufficiently remote from sulfides to be unaffected by the mineralizing event. At the time, only the Main Zone was well defined. Since then, the several other zones have been discovered and drilled. It is now clear that all of the sampled dolostone sites are in the central part of a large hydrothermally altered and mineralized area (Figs. 2 and 3).

The origin of the sulfide deposits in the Nanisivik area is still not clear. Although the sulfides are obviously epigenetic, Olson (1984, 1986) and Ford (1986) have presented somewhat contrary views regarding the nature of sulfide emplacement. The arguments revolve around whether the sulfides simply filled pre-existing open caverns, as suggested by the geometry of the deposits, or whether the ore fluids themselves were sufficiently corrosive to effect carbonate dissolution during the overall process of sulfide precipitation, as suggested by the mineral zoning of micro to macro scale found in the deposits. Based on geothermal gradients calculated from fluid inclusion homogenization temperatures for the ore stage minerals, McNaughton and Smith (1986) estimated that the minimum depth of ore emplacement was ~1600 m.

Relative age of mineralization

The ore was clearly emplaced later than deposition of the Society Cliffs Formation dolostones and later than local tilting of these strata, as both the tabular and keel bodies are discordant to the bedding. Moreover, the ore is younger than the Victor Bay Formation as the Main Zone passes through the Keystone Fault, which has dropped the formation by 150 m (Sutherland and Dumka 1995). South of the fault, the ore is situated within the Society Cliffs Formation just beneath the contact with the overlying Victor Bay shales (Fig. 2). In the same general area, the Main Zone is cut by the Mine Dike that cuts the Proterozoic rocks but not the Paleozoic formations. Thus, the Nanisivik sulfide deposits were emplaced sometime in the Proterozoic during the interval between deposition of the Victor Bay Formation and intrusion of the diabase dike. If the ≥1600 m depth estimate of McNaughton and Smith (1986) is correct, this would restrict Nanisivik ore deposition to a time no older than deposition of the overlying Elwin Formation (Table 1).

Thermal constraints

Fluid inclusion data have been reported by Jalonen (1982), Olson (1984) and McNaughton and Smith (1986). The latter reported homogenization temperatures for 222 inclusions in sphalerite and 135 inclusions in co-existing sparry dolomite from 23 samples collected throughout the Main Zone at Nanisivik. They found that the fluid inclusion homogenization temperatures for sphalerite and dolomite were statistically the same for both minerals in any given sample. Except for two samples collected from beside the Mine Dike that yielded an elevated average homogenization temperature of 286 ± 30 °C, the remaining samples gave a mean temperature of 163 ± 38 °C (1σ limits) with a range from 103 to 232 °C. In addition, the values showed a marked vertical gradient of ~60 °C/100 m in the ore zone, suggesting that the ore minerals were precipitated from ascending hydrothermal fluids. Further, from petrographic and geochemical analyses of the Society Cliffs dolostones some tens of kilometres away from the deposits, Ghazban et al. (1992) found that massive dolomitization of the precursor carbonates occurred in response to thermal convection of seawater beneath the overlying Victor Bay shale. This would have occurred prior to the precipitation of sparry dolomite such as that found in the ore.

Four lines of evidence indicate that the ore zones were not heated above 163 ± 38 °C since their deposition. First, even in the lowest formations of the Bylot Supergroup, the metamorphic levels are subgreenschist facies, reaching only prehnite–pumpelleyite grade in the Nauyat Formation (Jackson and Morgan 1978). Second, five illite crystallinity values from the Elwin Formation are characteristic of diagenetic temperatures rather than hotter metamorphic conditions (Knight and Jackson 1994). Third, the presence of diagenetic kaolinite in the supergroup (Knight and Jackson 1994) indicates that regional temperatures did not exceed 150 °C (Aldahan and Morad 1986). Fourth, the co-existence of wurtzite with pyrite in the ore indicates a maximum temperature for it of 155 to < 200 °C (Olson 1984).

Paleomagnetic methods

Three or four block samples, oriented using a magnetic compass, were collected at each of the 16 sites (Fig. 1; Table 2). They included 12 sites in the Society Cliffs dolostones, 2 in the 40-m-wide vertical Mine Dike, and 2 in the dolostones next to the dike that were collected for a paleomagnetic contact test (Everitt and Clegg 1962). Three or four specimens were drilled from each block so that most sites are represented by 12 specimens (Table 2). After storage for 2 months in a magnetically shielded room with an ambient field of ~100 γ in order to allow viscous remanence components to decay, the natural remanent magnetizations (NRM) of the specimens were measured using an automated Canadian Thin Films CTF-1 cryogenic magnetometer with a noise level of $< 4 \times 10^{-6}$ A/m. All subsequent measurements were done in the room.

Initially, two pilot specimens from each site were AF demagnetized in 12 steps up to 60 mT for specimens with the weakest NRM intensities or up to 120 mT for those with strong intensities using a Sapphire Instruments SI-4 AF demagnetizer. A third specimen from each site was thermally demagnetized in 12 steps up to 600 °C using a Schonstedt TSD-1 demagnetizer. Finding

Table 2 Site and unit mean remanence directions. Rock type: *D* dolostone; *Dc* dolostone in contact zone; *G* gabbroic dike. Samples: *B* blocks; *M* measured specimens; *E* end points used for site mean. Minimum (*Min.*) and Maximum (*Max.*) step temperature (°C) and coercivities (mT) included within the site mean. Direction: *Dec.* declination; *Inc.* inclination; α_{95} radius of cone of 95% confidence in degrees; *k* precision parameter of Fisher (1953)

Site	Rock type	Attitude		Median NRM intensity A/m	Samples			Demagnetization				Mean direction			
		Strike°	Dip°		B	M	E	Thermal		AF		Dec. °	Inc. °	α_{95} °	*k*
								Min. °C	Max. °C	Min. mT	Max. mT				
1	G	325	90	7.3 E-1	3	12	5[a]	–	–	15	90	106.0	39.6	18.3	18.4
2	Dc	260	15	5.9 E-4	3	13	12[a]	–	–	15	120	313.2	38.6	16.1	8.3
3	D	260	15	9.6 E-5	3	10	6[a]	–	–	15	120	296.9	55.2	17.9	14.9
12	D	260	15	3.5 E-5	4	12	10	300	400	10	40	291.1	62.4	13.7	13.5
16	D	260	15	9.3 E-5	3	12	12	200	500	15	60	299.1	68.5	5.4	64.4
18	D	260	15	5.6 E-5	3	12	12	300	400	15	60	310.4	47.3	12.2	13.6
19	G	325	90	6.8 E-2	4	12	7[a]	–	–	20	120	121.0	40.5	7.1	73.8
20	Dc	260	16	2.8 E-4	3	10	10	350	450	20	120	296.8	37.4	5.8	71.3
21	D	261	15	3.0 E-5	3	12	12[a]	300	400	15	60	307.2	57.6	10.3	18.6
22	D	259	18	4.5 E-5	3	12	12	350	500	20	60	299.3	51.2	6.4	46.9
23	D	303	17	4.6 E-5	3	12	8	–	–	15	60	296.3	57.4	9.9	32.0
24	D	229	16	3.1 E-5	3	12	9	300	450	15	40	308.9	53.8	14.5	13.5
25	D	262	10	2.3 E-5	3	12	0	–	–	–	–	–	–	–	–
26	D	262	10	3.1 E-5	3	12	0	–	–	–	–	–	–	–	–
27	D	194	4	7.8 E-5	3	12	12	300	450	15	80	296.5	44.6	5.9	54.3
28	D	266	16	4.0 E-5	3	12	12	300	500	20	60	293.2	53.3	7.4	35.4
1 + 19	G	–	–	–	6	24	12[a]	–	–	–	–	114.9	44.4	7.9	30.9
2 + 20	Dc	–	–	–	6	25	24	–	–	–	–	305.2	38.3	8.9	13.2
Mean	D	–	–	–	10[b]	–	–	–	–	–	–	300.2	55.3	4.8	104.1

[a] Site mean directions excluded to calculate the unit mean (Mean) direction
[b] Site mean directions used to average to calculate the unit mean (Mean) direction

that AF demagnetization was the more effective at isolating the characteristic remanent magnetization (ChRM) direction, the remaining specimens were demagnetized in seven AF steps in the optimal range indicated by the pilot specimens for each site. In addition, five dolostone and two dike specimens were subjected to saturation isothermal remanent magnetization (SIRM) analysis by magnetizing them in 12 direct-field steps up to 900 mT using a Sapphire Instrument SI-6 pulse magnetizer, and then by AF demagnetizing the SIRM in six steps to 40 mT. The ChRM directions were determined using the principal component method of Kirschvink (1980) and the site mean statistics using Fisher (1953).

Paleomagnetic results

Demagnetization and saturation remanence

The Society Cliffs dolostone specimens are weakly magnetized with a median NRM intensity of 5.3×10^{-5} A/m and with almost all specimens in the 10^{-5} A/m range, which is the practical limit of resolution for most cryogenic magnetometers. AF step demagnetization removed the viscous remanence by ~15 to 20 mT for most specimens and isolated the ChRM between ~20 and 60 mT, with intensities in the low 10^{-5} to high 10^{-6} A/m range before dropping below the $\sim 3 \times 10^{-6}$ A/m limit of resolution of the magnetometer (Fig. 5). The resulting ChRM directions for most dolostone specimens were directed downwards to the west-northwest at intermediate inclinations and they were reasonably well clustered for most sites (Table 2). This direction is termed the A ChRM component. Two of the 12 sites (sites 25 and 26) proved to be either too weakly magnetized to retain a measurable ChRM or magnetized with an antiparallel upwards ChRM that could not be distinguished from the downwards viscous remanence. The latter appears likely because the remanences of several specimens were rotating to a reversed polarity on AF step demagnetization. None of the dolostone specimens show a significant NRM intensity drop in the 275 to 325 °C unblocking temperature range that is diagnostic of pyrrhotite, indicating it is not a significant carrier of the ChRM. However, several specimens carry the A component direction through much of the 500 to 585 °C range, which is indicative of either magnetite or low-titanium titanomagnetite (Fig. 6).

Using the magnetic relaxation time versus blocking temperature curves of Pullaiah et al. (1975; see also Dunlop 1995) for magnetite and assuming a higher-than-likely precipitation temperature for the Nanisivik ore of about 200 °C (i.e. ~163° + 38 °C) that lasted for a longer-than-likely precipitation period of about 10 m.y., the measured ChRMs residing in magnetite at blocking temperatures above 375 °C would certainly record the primary precipitation of magnetite and sparry dolomite with sulfides in the dolostones and not some later secondary remagnetization event.

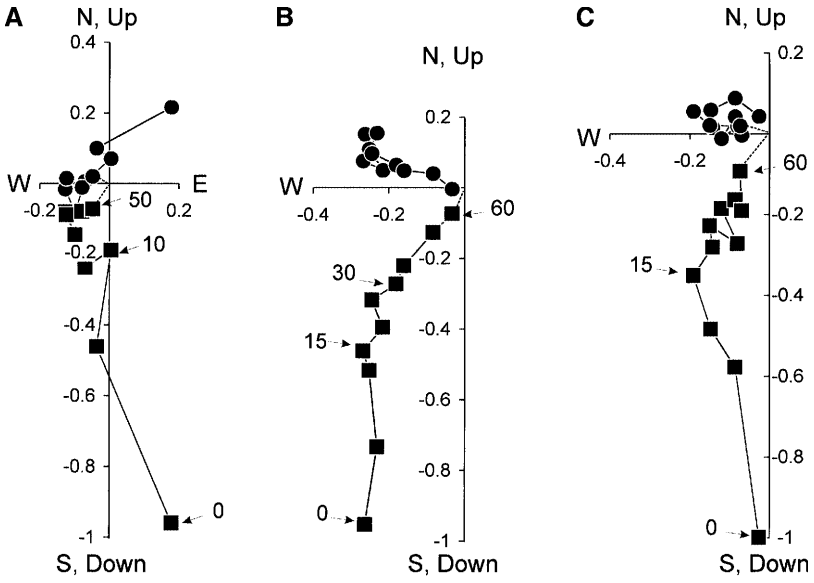


Fig. 5 Orthogonal vector component plots on AF step demagnetization of example dolostone specimens from: **A** site 12, $J_0 = 4.01 \times 10^{-5}$ A/m; **B** site 18, $J_0 = 5.26 \times 10^{-5}$ A/m; and **C** site 23, $J_0 = 5.73 \times 10^{-5}$ A/m. The axes are: *N* north, *E* east, *S* south, *W* west. The axial units are expressed as a ratio of the natural remanent magnetization intensity (J_0). Solid *circles* (*squares*) indicate points in the horizontal (vertical) plane. Some AF steps are labelled in millitesla

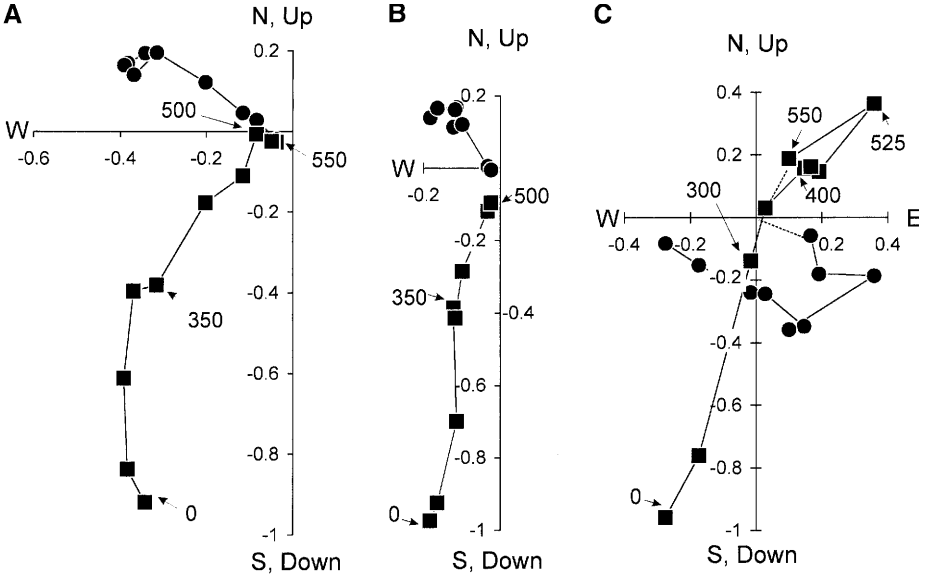


Fig. 6 Orthogonal vector component plots on thermal step demagnetization of example dolostone specimens from: **A** site 16, $J_0 = 19.2 \times 10^{-5}$ A/m; **B** site 26, $J_0 = 3.06 \times 10^{-5}$ A/m; and **C** site 28, $J_0 = 13.7 \times 10^{-5}$ A/m. Conventions as in Fig. 5 except that some steps are labelled in °C

Given that the SIRM showed a rapid acquisition of magnetization up to 350 mT and then levelled off at near-saturation (Fig. 7A), a pattern that indicates the presence of either pyrrhotite or magnetite, the obvious conclusion is that the A ChRM is carried by magnetite or low-titanium titanomagnetite. The slow climb from ~350 mT up to the maximum field of 900 mT indicates the presence of a minor amount of goethite or hematite, which are secondary minerals in the paragenetic sequence due to weathering (Olson 1984). Further, AF demagnetization of the SIRM showed that the magnetite carriers are mostly single domain to pseudosingle domain, a fact which further favours their retention of a geologically significant ChRM (Fig. 7B).

When the mean A ChRM directions of the dolostones are averaged for the ten sites without a bedding tilt correction, they give a unit mean declination (*D*) of 300.2°, an inclination (*I*) of 55.4°, radius of the cone of 95% confidence (α_{95}) of 4.8°, and a precision parameter (*k*) of 104.1 where the number of sites (*N*) is 10 (Fisher 1953; Fig. 8A).

Fold test and pole positions

A paleomagnetic fold test (Graham 1949) was run on the A site mean ChRM directions. After applying a tilt correction, the unit mean direction of A becomes $D = 312.1°$, $I = 45.4°$, $k = 76.6$ and $\alpha_{95} = 5.6$ for $N = 10$. The result shows that tilt corrections cause the directions to scatter, resulting in *k* decreasing from 104.1 to 76.6. Although this decrease is not statistically significant at the 95% confidence level (McFadden and Jones 1981), it is strongly indicative that the remanence

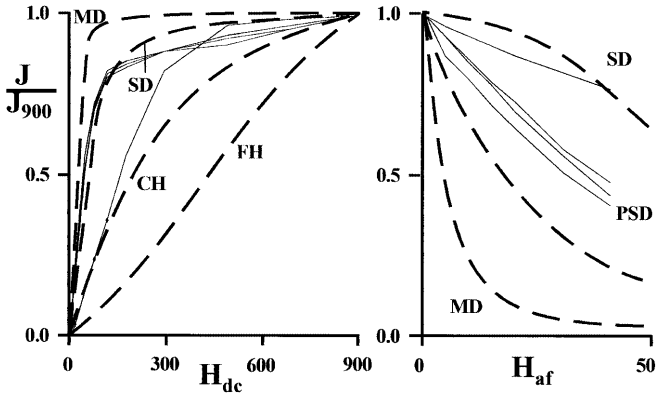


Fig. 7 Saturation isothermal remanent magnetization (SIRM) plots for example specimens showing: **A** acquisition of SIRM and **B** AF demagnetization of SIRM. J/J_{900} is the ratio of the measured remanence intensity to the SIRM intensity at 900 mT. H_{dc} is the magnetizing direct and H_{af} is the demagnetizing alternating field intensity in millitesla. The type curves bound the ranges for single (*SD*), pseudosingle (*PSD*) and multidomain (*MD*) magnetite, and for coarse- (*CH*) and fine-grained (*FH*) hematite

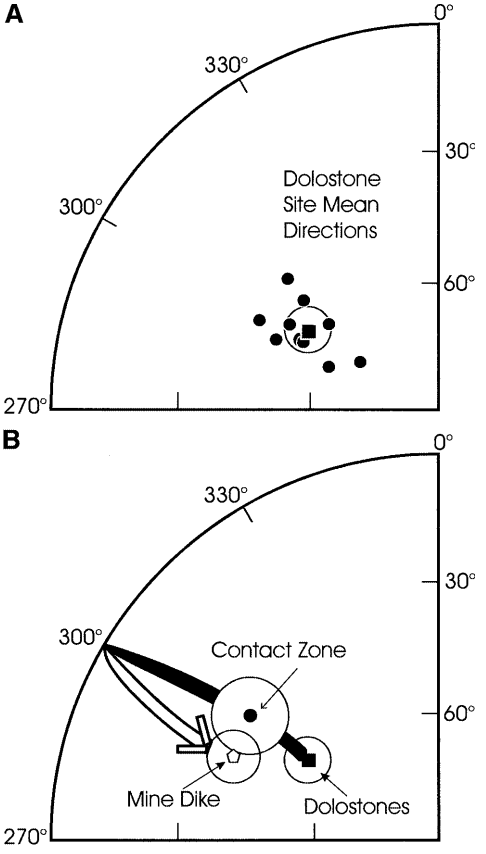


Fig. 8 Northwest quadrant of an equal-area stereogram showing: **A** the dolostone site mean ChRM directions (*circles*) with their mean (*square*) and cone of 95% confidence, and **B** the mean directions for the dolostones, for the dolostones in the altered zone next to the Mine Dike, and for the Mine Dike itself. *Solid (open) symbols* denote downward (upward) directions in the lower (upper) hemisphere

was acquired after tilting of the carbonate beds. Therefore, using the uncorrected unit mean direction for A, the resulting pole position is at 168.2°E longitude and 42.8°N latitude, with radii for the oval of 95% confidence of $\delta_p = 4.9°$ and $\delta_m = 6.8°$. This pole position with its paleolatitude of 35°S indicates that these host dolostones, deposited at ~1250 Ma and dolomitized at least in part at ~1200 ± 20 Ma, were remagnetized at ~1095 ± 10 Ma (Fig. 4). This remagnetization could be caused by: (1) original limestones being chemically altered by magnesium substitution to form dolostones, or (2) more likely, original dolostones being recrystallized with coincident precipitation of minute amounts of magnetite, probably as inclusions in the dolomite grains.

Paleomagnetic contact test

The Mine Dike cuts the Main Zone at Nanisivik (Fig. 2), altering the sulfides and host dolostones at its contacts (Sutherland and Dumka 1995). At the mine the dike is termed a "gabbro" because of its coarse crystallinity. It trends ~N25°W, or about the same trend as the so-called Borden dikes on the Borden Peninsula. This ~40-m-wide diabase dike has an NRM intensity of $\sim 1 \times 10^{-1}$ A/m. Thermal demagnetization and SIRM analysis show that the NRM is carried mostly by multidomain titanomagnetite with large viscous components that make its ChRM difficult to isolate. Nevertheless, several specimens show similar directions of both polarities at both sites 1 and 19 (Table 2) and combine to give a mean of $D = 114.9°$, $I = 40.4°$, $\alpha_{95} = 7.9°$ and $k = 30.9$ where the number of specimens is 12 (Fig. 4). This B component direction is statistically significantly different from the A direction (McFadden and Lowes 1981), and it gives a virtual pole position at 155.3°E, 15.1°S ($\delta_p = 5.8°$, $\delta_m = 9.5°$). A comparison with the poles obtained by Christie and Fahrig (1983) from diabase dikes elsewhere on Baffin Island indicates that they would have considered the Mine Dike to belong to the so-called "~950 Ma Borden" dike swarm (Fig. 4).

Samples from sites 2 and 20 were collected from Society Cliffs dolostones within 0.2 to 2.0 m of the Mine Dike respectively. These samples are clearly altered, giving NRM intensities of $\sim 4 \times 10^{-4}$ A/m or about one order of magnitude greater than the dolostones elsewhere (Table 2). Thermal demagnetization and SIRM analysis indicate that a majority of the NRM in the altered zone is carried by hematite. Both sites 2 and 20 give similar ChRM directions with mixed polarities (Table 2), but have differing distributions. At site 20 the directions have a circular distribution about the mean, giving good cluster statistics. However, at site 2, the directions extend along a great circle that starts close to the dolostone direction of $D = 300°$, $I = 55°$, continues close to the Franklin dike direction of $D = 305°$, $I = -10°$, and extrapolates beyond it to close to the antipodal "Borden" dike direction of $D = 300°$, $I = -50°$ (Fig. 6B). Regardless of which dike direction is

primary, the results provide a paleomagnetic contact test that shows that the ChRM of the dolostones in the mineralized zone at Nanisivik predates the ChRM of the ~720 Ma Mine Dike (Everitt and Clegg 1962).

Discussion

Age of the dolostones remagnetization

The apparent polar wander path (APWP) for the North American craton is relatively well known for the time interval between ~1270 and ~720 Ma. Figure 4 shows this path, assembled from several sources as noted in the caption. The significant result is that the pole position of the dolostones overlaps onto the well-defined "Logan Loop" portion of the path at ~1095 ± 10 Ma at the start of a predominantly normal polarity interval. Noting that this portion of the path is defined almost entirely by about 30 poles from 1110 to 1085 Ma rock units in the Keweenawan rocks around Lake Superior, the offset of the dolostones pole from the path, which is statistically non-significant, probably reflects minor relative rotation at most between the two areas after ~1100 Ma. Three other points should be made. First, the dolostones' pole position falls >20° of arc away from any part of the North American APWP path after ~700 Ma (Dunlop et al. 1985; Van der Voo 1990; Besse and Courtillot 1991). Second, the pole position is distinctly different from that given by any other observed grouping of ChRM directions for Baffin Island (Christie and Fahrig 1983; Pehrsson and Buchan 1999). Third, the dolostone pole falls in a logical succession along the APWP relative to other Bylot Supergroup and related poles (Fig. 4). Therefore, we conclude that the 1095 ± 10 Ma dolostone pole position dates the hydrothermal event that formed the mineralization at Nanisivik. Mineral magnetics tests to tie the remagnetization to the mineralization event, such as those demonstrated by Symons et al. (1996) for several other Mississippi Valley-type ore deposits, would be a useful addition to this research.

Possibility of a Phanerozoic remagnetization

Pehrsson and Buchan (1999) have recently argued that the "Borden" dike ChRM directions record Cretaceous–Early Tertiary partial remagnetizations rather than true ~720 Ma Franklin directions, but it is difficult to visualize how such an overprinting event could remagnetize the Mine Dike to a "Borden-like" direction (Fig. 2) without remagnetizing the host dolostones also. "Borden" ChRM directions are about $D = 300°$, $I = -50°$ if normal, or $D = 120°$, $I = 50°$ if reversed, whereas Franklin directions are about 40° away at about $D = 305°$, $I = -10°$ if normal, or $D = 125°$, $I = 10°$ if reversed. The Mine Dike and "Borden" dikes elsewhere have mixed polarities and, similarly, the Franklin dikes exhibit both polarities. Fahrig et al. (1971) and Fahrig and Schwarz (1973) attributed the "Borden" directions to partial remagnetization of Franklin directions. Christie and Fahrig (1983) concluded that the "Borden" dikes were a separate ~950 Ma swarm based on: (1) a slightly different directional trend of ~20°; (2) cross-cutting relationships between "Borden" and Franklin dikes; (3) a K–Ar age for a "Borden" dike of 819 ± 107 Ma that was notably older than the 623 ± 77 Ma average yielded by the Franklin dikes; (4) two paleomagnetic baked contact tests; and, (5) the fact that the mixed polarity of the "Borden" dikes is difficult to achieve by later remagnetization.

Pehrsson and Buchan (1999) resurrected the partial remagnetization argument because the "Borden" dikes give U–Pb baddeleyite ages of ~720 Ma. They suggested that the "Borden" dikes are older, but only by a couple of million years. This interpretation can explain: the geographic distribution of the "Borden" dikes in the west-central portion of Baffin Island, the slightly different structural trend of the two dike swarms, the observations of Franklin dikes cutting "Borden" dikes, and the positive, but contestable, paleomagnetic contact tests. They noted that the 819 ± 107 Ma K–Ar age on a "Borden" dike is correct within its error limits, and that anomalously old K–Ar ages can result from excess argon. To explain the "Borden" dike ChRM directions, Pehrsson and Buchan (1999) pointed out that the site poles for most "Borden" dikes lie on a great circle that passes through the true Franklin pole and the Cretaceous–early Tertiary 130–60 Ma section of the APWP, which is the only part of the North American path it intersects, and they suggested that tectonism associated with the opening of the Labrador Basin during the Cretaceous–early Tertiary (Trettin 1989; Chalmers 1991) was responsible for remagnetizing the "Borden" dikes by either thermal or chemical overprinting.

We disagree with Pehrsson and Buchan (1999). As noted previously, several lines of thermal evidence indicate that no thermal pulse has affected the dolostones at Nanisivik since mineralization except within the "Borden" Mine Dike contacts. Acceptance of a Cretaceous–early Tertiary chemical remagnetization would require that the dolostones' cluster of A ChRM directions be fortuitously different from "Borden" dike directions, and that most of the isotope geochemical systems in the dolostones similarly be unaffected (Ghazban et al. 1992; Kah et al. 1999). Further the Earth's magnetic field had a normal polarity for almost all of Cretaceous–early Tertiary time so that, while overprinting of Franklin reversed to a "Borden" reversed directions is tenable, overprinting of Franklin normal to "Borden" normal directions is not.

We suggest that the "Borden" dikes acquired their ChRM direction at ~720 Ma when the slightly later Franklin dikes were emplaced. This could happen if the "Borden" dikes were emplaced over a very short period of time so that the secular variation was not truly averaged out. Alternatively, they could have been

emplaced during a reversal or an intensity low of the Earth's magnetic field, a feature that is indicated by the bipolar nature of the ChRM for both the "Borden" dikes and Franklin dikes. Another possibility is that the "Borden" dikes, were derived from an earlier magmatic phase with a somewhat different ferromagnetic mineralogy than the later Franklin dikes, so that they are not fully demagnetized as easily by AF or thermal step demagnetization. Note also that most of the Baffin dike paleomagnetic studies were done using less rigorous demagnetization protocols than current norms. In short, there are a number of other possibilities to explain the aberrant "Borden" dike directions without invoking an endemic Cretaceous–early Tertiary remagnetization for the entire west-central Baffin Island region.

Ore genesis

The implications of a $\sim$1095 $\pm$ 10 Ma age for the genesis of the Nanisivik mineralization merit comment also. Paleomagnetic data and Pb–Pb ages show deposition and initial microdolomitization of the Society Cliffs dolostone by $\sim$1250 Ma and $\sim$1200 $\pm$ 20 Ma, respectively (Fahrig et al. 1981, as recalculated above). The Victor Bay shales and overlying strata of the Bylot Supergroup were then deposited, accompanied by syndepositional movement on the northwest–southeast faults that define the horsts and grabens of the rift system and by northward tilt of the strata. This allowed ascending hydrothermal fluids to move up the faults and precipitate the iron–zinc–lead sulfide mineralization and dolomite at 1095 $\pm$ 10 Ma, with the Victor Bay shales possibly acting as an aquitard. If the mineralization event had happened during deposition of the Strathcona Sound Formation that records the "Rift-II" event, as Knight and Jackson (1994) suggest, then it would require an extreme geothermal gradient of $\sim$100 °C/km above the base of the Victor Bay shale. Then also, the Strathcona Sound pole of Fahrig et al. (1981) should coincide with the dolostone pole found here, but it does not. More likely, therefore, is that the hydrothermal event occurred after deposition of the entire Bylot Supergroup when: (1) the required gradient would be a more reasonable $\sim$40°C/km; (2) the opportunity for basin dewatering by burial loading is maximized; and (3) differing pole positions for the Strathcona Sound Formation and Nanisivik Mine dolostones are given that fall in a logical progression along the APWP.

The source of the mineralizing fluids is speculative. An obvious possibility is that they were meteoric and entrapped waters that were driven downwards along the northwest–southeast rift faults out of an uplifted block(s), heated at depth, and then moved up faults into the mineralized block. Another possibility is that there was further rifting along the same trend that provided the hydrothermal fluids, but that the geologic record of the event disappeared during the long hiatus from $\sim$1095 Ma to emplacement of the Franklin dikes at $\sim$720 Ma.

Conclusions

Radiometric and paleomagnetic data indicate that the Society Cliffs dolostones were originally deposited by $\sim$1250 Ma and regionally dolomitized at $\sim$1200 $\pm$ 20 Ma. Within the mineralized region around the Nanisivik sulfide deposits, the dolostones yield a stable ChRM direction that apparently postdates tilting of the strata, predates intrusion of the $\sim$720 Ma Franklin dikes and gives a pole position at 1095 $\pm$ 10 Ma. This A ChRM is deemed to be a chemical remagnetization upon recrystallization of the host dolostones that is coeval with the hydrothermal fluid flow event that precipitated the mineralization. Nanisivik is, to our knowledge, the first dated Precambrian MVT deposit. A valuable extension of this study would be to compare the paleomagnetic, petrologic and geochemical characteristics of Society Cliffs dolostones from sites that are truly remote from the Nanisivik sulfide zones to those of sites in the hydrothermal region and in the zones themselves.

Acknowledgements The authors thank: R. Sutherland, Senior Mine Geologist, D. Dumka, Mine Geologist, and Nanisivik Mines Ltd. for permission, help and geologic advice; J. Kulhanek who made the remanence measurements; M.T. Cioppa for technical and drafting assistance; K.L. Buchan and an anonymous referee for helpful reviews that materially improved this paper; and the Natural Sciences and Engineering Research Council of Canada and the Geological Survey of Canada for funding this research.

References

Aldahan AA, Morad S (1986) Mineralogy and chemistry of diagenetic clay minerals in Proterozoic sandstones from Sweden. Am J Sci 286: 29–80

Besse J, Courtillot V (1991) Revised and synthetic apparent polar wander paths of the African, Eurasian, North American and Indian plates, and true polar wander since 200 Ma. J Geophys Res 96: 4029–4050

Chalmers JA (1991) New evidence on the structure of the Labrador Sea/Greenland continental margin. J Geol Soc Lond 148: 899–908

Christie KW, Fahrig WF (1983) Paleomagnetism of the Borden dykes of Baffin Island and its bearing on the Grenville loop. Can J Earth Sci 20: 275–289

Dunlop DJ (1995) Magnetism in rocks. J Geophys Res 100: 2161–2174

Dunlop DJ, Hyodo H, Knight T, Steele AG (1985) Paleomagnetism of the Tudor gabbro, Ontario: evidence for divergence between Grenvillia and interior Laurentia. R Astron Soc, Geophys J 83: 699–720

Ernst RE, Buchan KL (1993) Paleomagnetism of the Abitibi dyke swarm, southern Superior Province, and implications for the Logan Loop. Can J Earth Sci 30: 1886–1897

Everitt CWF, Clegg JA (1962) A field test of paleomagnetic stability. Am Assoc Petrol Geol Bull 41: 2120–2127

Fahrig WF, Schwarz EJ (1973) Additional paleomagnetic data on the Baffin diabase dikes and a revised Franklin pole. Can J Earth Sci 10: 576–581

Fahrig WF, West TD (1986) Diabase dyke swarms of the Canadian Shield. Geol Surv Can Map 1627 A

Fahrig WF, Irving E, Jackson GD (1971) Paleomagnetism of the Franklin diabases. Can J Earth Sci 8: 455–467

Fahrig WF, Christie KW, Jones DL (1981) Paleomagnetism of the Bylot basins: evidence for Mackenzie continental tensional tectonics. In: Campbell FHA (ed) Proterozoic basins of Canada. Geol Surv Can Pap 81-10, pp 303–312

Fisher RA (1953) Dispersion on a sphere. R Astron Soc Lond Proc A217: 295–305

Ford DC (1986) Genesis of paleokarst and strata-bound zinc–lead deposits in a Proterozoic dolostone, northern Baffin Island, Canada – a discussion. Econ Geol 81: 1562–1563

Ghazban F, Schwarcz HP, Ford DC (1992) Multistage dolomitization in the Society Cliffs Formation, northern Baffin Island, N.W.T., Canada. Can J Earth Sci 29: 1459–1473

Graham JW (1949) The stability and significance of magnetism in sedimentary rocks. J Geophys Res 54: 131–167

Heaman LM, LeCheminant AN, Rainbird RH (1992) Nature and timing of Franklin igneous events, Canada: implications for a Late Proterozoic mantle plume and the break-up of Laurentia. Earth Planet Sci Lett 109: 117–131

Irving E, Park JK, McGlynn JC (1972) Paleomagnetism of the Et-Then Group and Mackenzie diabase in the Great Slave area. Can J Earth Sci 9: 744–755

Jackson GD, Iannelli TR (1981) Rift-related cyclic sedimentation in the Neohelikian Borden Basin, northern Baffin Island. In: Campbell FHA (ed) Proterozoic basins of Canada. Geol Surv Can Pap 81-10, pp 269–302

Jackson GD, Morgan WC (1978) Precambrian metamorphism on Baffin and Bylot Islands. In: Fraser JA, Heywood WW (eds) Metamorphism in the Canadian Shield. Geol Surv Can Pap 78-10, pp 249–267

Jackson GD, Hunt PA, Loveridge WD, Parrish RR (1990) Reconnaissance geochronology of Baffin Island, N.W.T. In: Radiogenic age and isotopic studies: report 3. Geol Surv Can Pap 89-2, pp 123–148

Jalonen MR (1982) Fluid inclusion studies on the Nanisivik zinc–lead deposit, N.W.T. B.Sc. Thesis, Univ Windsor

Kah LC, Sherman LG, Narbonne GM, Knolf AH, Kaufmann AJ (1999) $\delta^{13}C$ stratigraphy of the Proterozoic Bylot Supergroup, Baffin Island, Canada: implications for regional lithostratigraphic correlations. Can J Earth Sci 36: 313–332

Kirschvink JL (1980) The least-squares line and plane and the analysis of paleomagnetic data. R Astron Soc, Geophys J 62: 699–718

Knight RD, Jackson GD (1994) Sedimentology and stratigraphy of the Mesoproterozoic Elwin Subgroup (Aquigilik and Sinasiuvik Formations), uppermost Bylot Supergroup, Borden Rift Basin, Northern Baffin Island. Geol Surv Can Bull 45

LeCheminant AN, Heaman LM (1989) Mackenzie igneous events, Canada: Middle Proterozoic hotspot magmatism associated with ocean opening. Earth Planet Sci Lett 96: 38–48

LeCheminant AN, Heaman LM (1991) U–Pb ages for the 1.27 Ga Mackenzie igneous events, Canada; support for a plume initiation model. Geol Assoc Can Abstr 16: A73

McFadden PL, Jones DL (1981) The fold test in palaeomagnetism. R Astron Soc, Geophys J 67: 53–58

McFadden PL, Lowes FJ (1981) The discrimination of mean directions drawn from Fisher distributions. R Astron Soc, Geophys J 67: 19–33

McNaughton K, Smith TE (1986) A fluid inclusion study of sphalerite and dolomite from the Nanisivik lead–zinc deposit, Baffin Island, Northwest Territories, Canada. Econ Geol 81: 713–720

Olson RA (1984) Genesis of paleokarst and strata-bound zinc–lead sulphide deposits in a Proterozoic dolostone, Northern Baffin Island, Canada. Econ Geol 79: 1056–1103

Olson RA (1986) Genesis of paleokarst and strata-bound zinc–lead deposits in a Proterozoic dolostone, northern Baffin Island, Canada – a reply. Econ Geol 81: 1563–1566

Palmer HC, Hayatsu A (1975) Paleomagnetism and K–Ar dating of some Franklin lavas and diabases, Victoria Island. Can J Earth Sci 12: 1439–1447

Pehrsson SJ, Buchan KL (1999) Borden dykes of Baffin Island, N.W.T.: a Franklin U–Pb baddeleyite age and a paleomagnetic reinterpretation. Can J Earth Sci 36: 65–73

Pullaiah G, Irving E, Buchan KL, Dunlop DJ (1975) Magnetization changes caused by burial and uplift. Earth Planet Sci Lett 28: 133–143

Sutherland RA, Dumka D (1995) Geology of the Nanisivik Mine, N.W.T., Canada. Soc Econ Geol Guideb Ser. 22: 4–12

Symons DTA, Lewchuk MT, Dunlop DJ, Constanzo-Alverez V, Halls HC, Bates MP, Palmer HC, Vandall TA (1994) Synopsis of paleomagnetic studies in the Kapuskasing structural zone. Can J Earth Sci 31: 1206–1277

Symons DTA, Sangster DF, Leach DL (1996) Palaeomagnetic dating of Mississippi Valley-type Pb–Zn–Ba deposits. Soc Econ Geol, Spec Pub No. 4: 515–526

Trettin HP (1989) The Arctic islands. In: Bally AW, Palmer AR (eds) The geology of North America – an overview. Geol Soc Am, The Geology of North America 4, pp 349–370

Van der Voo R (1990) Phanerozoic paleomagnetic poles from Europe and North America and comparisons with continental reconstructions. Rev Geophys 28: 167–206